

MUHAMMAD RESHAEEL, PhD

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RESEARCH PROFILE

I hold a PhD in Mechanical Engineering with expertise in building energy systems, HVAC systems, and advanced thermal energy systems. My research develops physics-informed and data-driven models that combine thermodynamic principles, experimental data, and machine/deep learning for energy prediction, system optimization, and fault detection and diagnostics. I am particularly interested in extending this background toward building energy simulation, HVAC/control-oriented modeling, and AI-enabled optimization tools, with experience in Python, MATLAB-based modeling, and Modelica.

RESEARCH INTEREST

Physics-informed machine and deep learning for energy system modeling | Energy prediction | Fault detection and diagnostics | Modelling and optimization of HVAC systems, buildings, and advanced power systems | Physics-guided and data-driven analysis of thermodynamic behavior of energy systems | Waste heat recovery and repowering of power systems | Building energy performance and system-level energy analysis

EDUCATION

Khalifa University, Abu Dhabi, UAE PhD in Mechanical Engineering Thesis: Advanced Cooling Technology for Building Energy Efficiency.	Aug 2022- May 2026 CGPA: 3.79/4.00
National University of Sciences and Technology (NUST), Islamabad, Pakistan Master of Science in Thermal Energy Engineering Thesis: Parametric Optimization of a Double Stage Reheated Organic Rankine Cycle (ORC) for Recovering Waste Heat from a Combined Cycle Gas Turbine Power Plant in the Perspective of Repowering.	Sep 2019- Nov 2021 CGPA: 4.00/4.00 (Gold medalist)
University of Engineering and Technology, Peshawar, Pakistan Bachelors of Science in Mechanical Engineering Thesis: Comparative Study of Carbon Capturing Techniques with Design and Fabrication of a Prototype Model.	Sep 2015- Sep 2019 CGPA: 3.65/4.00 (With distinction)

RESEARCH EXPERIENCE

Thermal Power Lab, Khalifa University, Abu Dhabi, UAE Funding agency: ASPIRE-(Under VRI Funding) Position: Graduate Research Assistant Key projects: - Development of a test bench of variable refrigerant flow (VRF)-based HVAC system. - Energy modeling of an eco-villa located in Masdar city, Abu Dhabi using Hourly Analysis Program (HAP) and development of a preliminary Modelica model of a basic thermal/HVAC system to strengthen familiarity with equation-based building energy simulation. - Development of an accurate, robust, and interpretable grey-box energy prediction model of a VRF system using machine learning, experimental data and physics. - Development of a noise resilient-physics-guided (NRPG) hybrid machine learning fault detection and diagnosis (FDD) framework for reliable operation of VRF systems in smart buildings. - Open-set deployable hybrid machine learning FDD framework capable of classifying known faults while rejecting unfamiliar faulty conditions.	Aug 2022- May 2026
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Position: Graduate Researcher

Key projects:

- Real-time data collection of a 1600 MW combined cycle gas turbine (CCGT) powerplant located in Kot Addu.
- Simulation of the 1600 MW CCGT powerplant integrated with different configurations of organic Rankine cycle (ORC) operating on three different working fluids in Cycle Tempo software.
- Investigation of the impact of integrating ORC system with the CCGT unit in terms of repowering, waste heat recovery, GHG emissions, and specific water consumption.

LIST OF PUBLICATIONS

[1] Reshaeel, M., and Ali, M. I. H., A Hybrid Autoencoder–Random Forest Framework for Open-Set Recognition of Sensor and Communication Faults in a Variable Refrigerant Flow System with Real-Time Deployment Analysis, Energy, 2026.
<https://doi.org/10.1016/j.energy.2026.141403>

Top 1% (IF 9.4)

[2] Reshaeel, M., Khadkikar, V., and Ali, M. I. H., Physics-guided, noise-resilient and interpretable machine learning framework for fault detection and diagnosis in variable refrigerant flow systems, Energy and AI, 2026.
<https://doi.org/10.1016/j.egyai.2026.100766>

Top 4% (IF 9.6)

[3] Reshaeel, M., and Ali, M. I. H., Enhancing energy prediction in variable refrigerant flow cooling systems: A hybrid grey-box model integrating thermodynamic knowledge, experimental data and machine learning, Energy, 2026.
<https://doi.org/10.1016/j.energy.2025.139895>

Top 1% (IF 9.4)

[4] M.I. Khan, M. Reshaeel, F. Asfand, S.G. Al-Ghamdi, M. Farooq, M. Khan, et al., Concentrated solar power driven desalination systems: A techno-economic review, Renewable and Sustainable Energy Reviews, 2026.
<https://doi.org/10.1016/j.rser.2025.116311>

Top 2% (IF 16.3)

[5] Reshaeel, M., and Ali, M. I. H., Inherent thermodynamic performance assessment of a variable refrigerant flow system under transient cooling load, Case Studies in Thermal Engineering, 2025.
<https://doi.org/10.1016/j.csite.2025.106171>

Top 8% (IF 6.4)

[6] Reshaeel, M., Abdelsamie, M. M., and Ali, M. I. H., A critical review of the thermal-hydraulic performance of fin and tube heat exchangers using statistical analysis, International Journal of Thermofluids, 2024.
<https://doi.org/10.1016/j.ijft.2024.100858>

Top 10%

[7] Reshaeel, M., Javed, A., Jamil, A., Ali, M., Mahmood, M., and Waqas, A., Multiparametric optimization of a reheated organic Rankine cycle for waste heat recovery based repowering, Energy Conversion and Management, 2022.
<https://doi.org/10.1016/j.enconman.2022.115237> .

Top 2% (IF 10.9)

TEACHING AND INDUSTRIAL EXPERIENCE

Khalifa University, Abu Dhabi, UAE

Aug 2022- May 2026

Position: Graduate Teaching Assistant

Key Tasks:

- Taught undergraduate lab courses.
- Assisted in teaching undergraduate course of applied thermodynamics and fluid mechanics.
- Maintenance and troubleshooting of lab equipment.

University of Wah, Wah Cantt, Pakistan

Feb 2022 – Aug 2022

Position: Lecturer

Key Tasks:

- Taught undergraduate courses of thermodynamics, heat transfer and control theory.
- Preparation of the objective-based education (OBE) curriculum for the undergraduate courses in the mechanical engineering department.

CECOS University of IT and Emerging Sciences, Peshawar, Pakistan

Feb 2021 – Feb 2022

Position: Lab Engineer

Key Tasks:

- Taught undergraduate lab and theory courses.
- Maintenance and troubleshooting of powerplant lab equipment.
- Preparation of the objective-based education (OBE) curriculum for the undergraduate lab courses in the mechanical engineering department.

Fauji Fertilizers Bin Qasim Limited, Pakistan

Sep 2020 – Dec 2021

Position: Management Associate Engineer

Key Tasks:

- Mechanical maintenance of equipment.
- Cooling tower maintenance and analysis.

TECHNICAL EXPERTISE

Building Energy and HVAC

HAP | Modelica | HVAC/VRF system modeling | Building energy simulation | Cooling performance analysis

Machine Learning and AI

Python | MATLAB | Physics-informed machine learning | Grey-box modeling | Deep learning | Data pre-processing | Statistical analysis | Model validation

Fault Detection, Diagnostics, and Energy Prediction

Fault detection and diagnostics, FDD | Sensor and communication fault diagnosis | Open-set recognition | energy prediction | Interpretable ML | Noise-resilient modeling

Thermal Systems, Optimization, and Energy Analysis

Cycle-Tempo for powerplant modeling | Thermodynamics | Heat transfer | System-level energy and exergy analysis | Uncertainty quantification | Parametric optimization | Techno-economic analysis | GHG emission and water consumption analysis

Experimental and Laboratory Skills

HVAC test-bench development | Instrument calibration | Experimental data acquisition | Measurement validation

Additional Engineering Tools

SAM | ANSYS | Design-Expert

HONORS, AWARDS, AND CERTIFICATIONS

- **Graduate Research Award**

Awarding institute: Khalifa University

- **Doctoral Research Scholarship Award**

Awarding institute: Khalifa University

- **PRISM-NUST Merit Scholarship Award**

Awarding institute: National University of Sciences and Technology (NUST)

- **President Gold Medal Award in Master of Science in Thermal Energy Engineering**

Awarding institute: National University of Sciences and Technology (NUST)

- **Distinction in Bachelor of Science in Mechanical Engineering**

Awarding institute: University of Engineering and Technology (UET)

- **Best Final Year Project in Environment Category**

Awarding institute: University of Engineering and Technology (UET)

- **Benevolent Scholarship during Bachelors**

Awarding institute: Federal Government of Pakistan

- **Wind Energy Course**

Awarding institute: Technical University of Denmark (DTU)

- **Introduction to Programming with MATLAB**

Awarding institute: Vanderbilt University

LANGUAGES

Professional language: English speaking (advanced), reading (advanced), listening (advanced), writing (advanced) TOEFL iBT: 108/120
Mother tongue: Urdu

REFERENCES

Prof. Dr. Mohamed Ali Professor Department of Mechanical and Nuclear Engineering, Khalifa University Email: mohamed.ali@ku.ac.ae	PhD advisor
Prof. Dr. Dimitris Goussis Professor Department of Mechanical and Nuclear Engineering, Khalifa University Email: dimitris.goussis@ku.ac.ae	PhD research supervisory committee member
Dr. Majid Ali Associate Professor Thermal Energy Engineering, US-Pakistan Center for Advanced Studies in Energy (USPCAS-E), National University of Sciences and Technology (NUST) Email: majid@uspcase.nust.edu.pk	Co-advisor during Masters